# Report: DC Motor Control using Arduino UNO + L293D

## 1. Objective

To design, wire, and simulate a single DC motor control system using Arduino UNO and the L293D H-bridge motor driver. The system demonstrates bidirectional control (forward/backward) and speed control using PWM. This report documents the procedure, wiring with color codes, pin-to-pin mapping, and the mathematical model for PWM control.

## 2. Components

| Component | Quantity | Notes |
|---|---|---|
| Arduino UNO | 1 | Controller (5V logic) |
| L293D IC | 1 | H-bridge motor driver |
| DC Motor | 1 | 6–9V rated |
| Battery | 1 | 6–9V motor supply |
| Breadboard | 1 | For interconnections |
| Jumper Wires | As needed | Color-coded wiring |

## 3. Step-by-Step Procedure

1. Place the L293D across the breadboard center gap so pins on each side are isolated.
2. Connect common ground: Battery (–), Arduino GND, and L293D GND pins must share the same ground rail.
3. Provide logic power: Arduino 5V → L293D Pin 16 (Vcc1).
4. Provide motor power: Battery (+) → L293D Pin 8 (Vcc2).
5. Connect motor outputs: L293D Pins 3 & 6 → DC Motor terminals.
6. Connect control pins: IN1 → Arduino D8, IN2 → Arduino D9.
7. Enable pin for PWM speed: EN1 → Arduino D6 (PWM).
8. Upload the sketch and verify motion in simulation/Serial Plotter.

## 4. Wiring Map (Pin-to-Pin with Color Codes)

| Wire Color | From (Arduino / Battery) | To (L293D / Motor) | Purpose |
|---|---|---|---|
| Green | Arduino GND / Battery (–) | Pins 4, 5, 12, 13 (GND) | Common ground |
| Blue | Arduino 5V | Pin 16 (Vcc1) | Logic power |
| Red | Battery (+) | Pin 8 (Vcc2) | Motor supply |
| Orange | Arduino D8 | Pin 2 (IN1) | Direction control A |
| Orange | Arduino D9 | Pin 7 (IN2) | Direction control B |
| Purple | Arduino D6 (PWM) | Pin 1 (EN1) | Speed control (PWM) |
| Black | Motor lead | Pin 3 (OUT1) | Motor output |
| Black | Motor lead | Pin 6 (OUT2) | Motor output |

## 5. L293D Pin Map (Top View)

**Pin numbering (Top View):**

```
1 EN1 16 Vcc1 (Logic +5V)
2 IN1 15 IN4
```

```
3 OUT1 14 OUT4
4 GND 13 GND
5 GND 12 GND
6 OUT2 11 OUT3
7 IN2 10 IN3
8 Vcc2 9 EN2
```

## 6. Mathematical Model (PWM Speed Control)

Arduino PWM uses an 8-bit duty cycle. The analogWrite(value) maps value $\in$ [0,255] to duty cycle D = value/255. Motor voltage (average) approximates: V_avg ≈ D × V_motor_supply. Example: For value = 180 and V_motor_supply = 9V, D = 180/255 ≈ 0.706 → V_avg ≈ 6.35V. Direction is set by IN1/IN2: (HIGH, LOW) → Forward; (LOW, HIGH) → Reverse; (LOW, LOW) → Stop; (HIGH, HIGH) → Brake.

## 7. Direction Truth Table

| IN1 | IN2 | Result |
|---|---|---|
| HIGH | LOW | Forward |
| LOW | HIGH | Backward |
| LOW | LOW | Stop (Coast) |
| HIGH | HIGH | Brake |

## 8. Safety & Notes

• Always keep common ground between Arduino, L293D, and battery.

• Do not feed 9V into Arduino 5V pin. Use VIN/DC jack only if powering Arduino externally.

• 9V block batteries are weak for motors; prefer AA packs or Li-ion for real hardware.

• Ensure EN1 is driven HIGH or PWM; otherwise the motor will not spin.